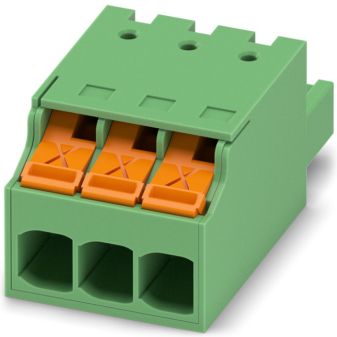
Data sheet

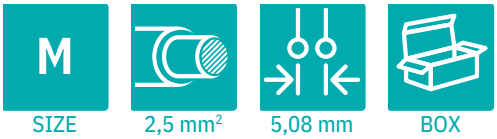
Item No.: 1491342

Type: XPC 2,5/ 3-ST-5,08

PCB connector, Push-X-connection



1 Main features



- No. of pos. 3
- Conductor cross section 2.5 mm²
- Color green (RAL 6021)
- Pitch 5.08 mm
- Connection method Push-X-connection
- Nominal current 16 A
- Nominal voltage 320 V
- Connection direction 0 °
- Type of packaging packed in cardboard

2 Your advantages

- Highly convenient operation thanks to direct, effortless, and tool-free conductor connection
- Quick connection of all conductor types with and even without ferrules
- Reliable wiring thanks to acoustic and visual feedback
- Quick conductor release by simply pressing the orange release button
- Plug-in compatibility with existing headers of the COMBICON portfolio
- Quick and convenient testing using integrated test option

Make sure you always use the latest documentation.
It can be downloaded at: phoenixcontact.com/product/1491342

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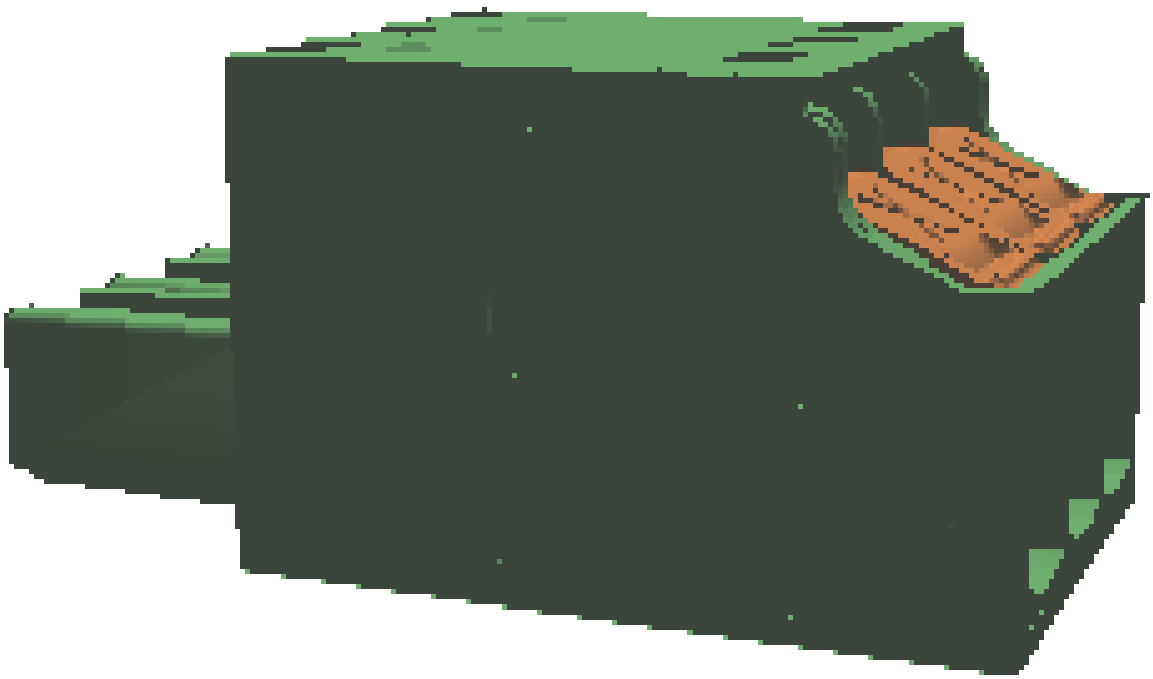
4 3D model in PDF can be activated (Acrobat Reader only)



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5 General Technical Data

5.1 item properties

Item no.	1491342
Type	XPC 2,5/ 3-ST-5,08
Product line	COMBICON Connectors M
Connector system	COMBICON MSTB 2,5
Product type	PCB connector
Contact connection type	Socket
Range of articles	XPC 2,5/..-ST
Pitch	5.08 mm
Number of positions	3
Number of rows	1
Connection method	Push-X-connection
Connection direction of the conductor to plug-in direction	0 °

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6 Mounting method

6.1 Flange mounting

Type of locking	without
Mounting flange	without

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7 Conductor connection

7.1 Connection capacity

Nominal cross section	2.5 mm^2
Conductor cross section, rigid	0.5 mm^2 ... 2.5 mm^2
Conductor cross section, flexible	0.5 mm^2 ... 2.5 mm^2
Conductor cross section flexible, with ferrule without plastic sleeve	0.5 mm^2 ... 2.5 mm^2
Conductor cross section flexible, with ferrule with plastic sleeve	0.5 mm^2 ... 2.5 mm^2
Cylindrical gauge a x b / diameter	2.8 mm x 2.0 mm / -
Stripping length	10 mm

7.2 Connection capacity AWG

Conductor cross section AWG	20 ... 12

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8 Material properties

8.1 Material of metal parts

Note	WEEE/RoHS-compliant, whisker-free acc. to IEC 60068-2-82/JEDEC JESD 201
Contact material	Cu alloy
Terminal point surface	Tin (4 - 8 µm Sn)
Surface contact area	Tin (4 - 8 µm Sn)
Surface characteristics	hot-dip tin-plated

8.2 Material of plastic parts

	Housing	Actuation element
Color	green (RAL 6021)	orange (RAL 2003)
Insulating material	PBT	PA
Insulating material group	II	I
CTI according to IEC 60112	575	600
Flammability rating according to UL 94	V0	V0

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9 Dimensions

9.1 Dimensions for the product

Length	30.86 mm
Width	16.86 mm
Total height	14.8 mm

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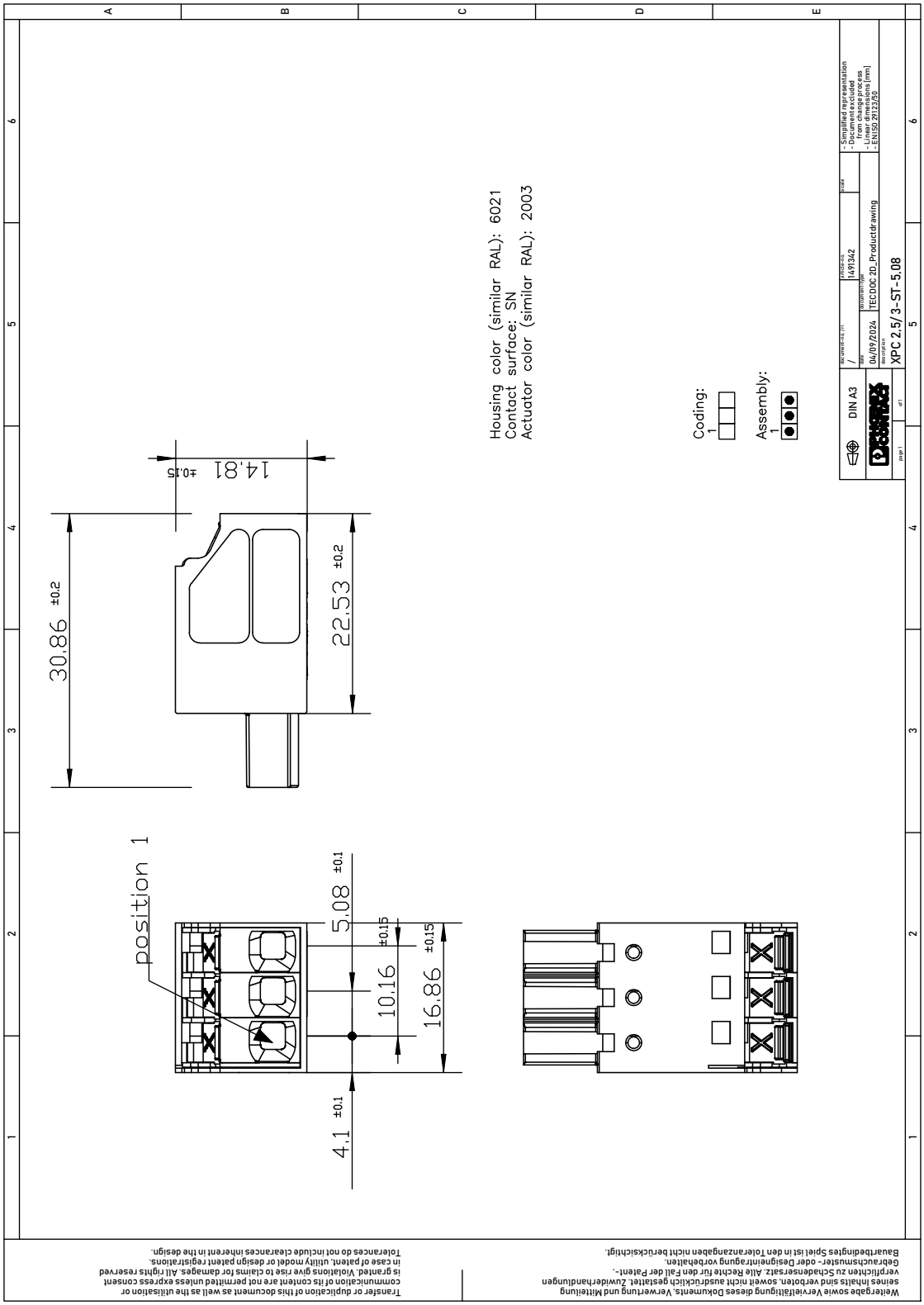
10 Product drawing



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11 Product notes

11.1 General information

Notes on operation	In accordance with IEC 61984, COMBICON connectors have no switching power (COC). During designated use, they must not be plugged in or disconnected when carrying voltage or under load.

12 Packaging specifications

Type of packaging	packed in cardboard
Packing unit	100

13 Application

Ferrules without insulating collar, according to DIN 46228–1	
Recommended crimping pliers	1212034 CRIMPFOX 6 1213144 CRIMPFOX CENTRUS 6S
Ferrules without insulating collar, according to DIN 46228–1	Cross section: 0.5 mm^2; Length: 10 mm Cross section: 0.75 mm^2; Length: 10 mm Cross section: 1 mm^2; Length: 10 mm Cross section: 1.5 mm^2; Length: 10 mm Cross section: 2.5 mm^2; Length: 10 mm

Ferrules with insulating collar, according to DIN 46228–4	
Recommended crimping pliers	1212034 CRIMPFOX 6 1213144 CRIMPFOX CENTRUS 6S
Ferrules with insulating collar, according to DIN 46228–4	Cross section: 0.5 mm^2; Length: 10 mm Cross section: 0.75 mm^2; Length: 10 mm Cross section: 1 mm^2; Length: 10 mm Cross section: 1.5 mm^2; Length: 10 mm Cross section: 2.5 mm^2; Length: 10 mm

13.1 Temperature limit values

Ambient temperature (storage/transport)	-40 °C ... 70 °C
Relative humidity (storage/transport)	30 % ... 70 %
Ambient temperature (assembly)	-5 °C ... 100 °C
Ambient temperature (operation)	-40 °C ... 105 °C (dependent on the derating curve)

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14 General tests

14.1 Specification

Specification	IEC 61984
Specification	IEC 60999-1
Brief description	PCB connector

15 Mechanical tests

15.1 Check for damage to conductor or loosening

Result	Test passed
Specification	IEC 60999-1:1999-11

15.2 Pull-out test

Specification	IEC 60999-1:1999-11
Result	Test passed
Conductor cross section/conductor type/tractive force actual value	0.5 mm^2 / solid / > 20 N
Conductor cross section/conductor type/tractive force actual value	0.5 mm^2 / flexible / > 20 N
Conductor cross section/conductor type/tractive force actual value	2.5 mm^2 / solid / > 50 N
Conductor cross section/conductor type/tractive force actual value	2.5 mm^2 / flexible / > 50 N

15.3 Repeated connection and disconnection

Specification	IEC 60999-1:1999-11
Result	Test passed

15.4 Conductor connection

Specification	IEC 60999-1:1999-11
Result	Test passed

15.5 Visual examination

Specification	IEC 61984:2008-10
Visual examination	Test passed
Specification	IEC 60512-1-1:2002-02

15.6 Dimensional test

Dimensional test	Test passed
Specification	IEC 60512-1-2:2002-02

15.7 Resistance of marking

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Resistance of marking	Test passed
Specification	IEC 60068-2-70:1995-12

15.8 Polarization and coding

Polarization when inserted Requirement >20 N	Test passed
Specification	IEC 60512-13-5:2006-02

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16 Insertion and withdrawal forces

Insertion and withdrawal force	Test passed
Specification	IEC 60512-13-2:2006-02
No. of cycles	25
Insertion strength per pos. approx.	9 N
Withdraw strength per pos. approx.	8 N

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17 Electrical tests

Rated current / conductor cross section	16 A / 2.5 mm^2
Rated insulation voltage (III/2)	320 V
Rated surge voltage (III/2)	4 kV
Contact resistance	1.2 mΩ
Degree of pollution	2

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18 Air and creepage distances

Component	PCB connector		
Specification	IEC 60664-1:2020-05		
Mains type	unearthed mains		
Insulating material group	II		
Comparative tracking index (IEC 60112)	CTI 575		
Rated insulation voltage	250 V	320 V	500 V
Rated surge voltage	4 kV	4 kV	4 kV
Degree of pollution	3	2	2
Overvoltage category	III	III	II
Minimum clearance case A (inhomogeneous field)	3 mm	3 mm	3 mm
Minimum value of the creepage path requirement in acc. with table	3.6 mm	3 mm	3.6 mm

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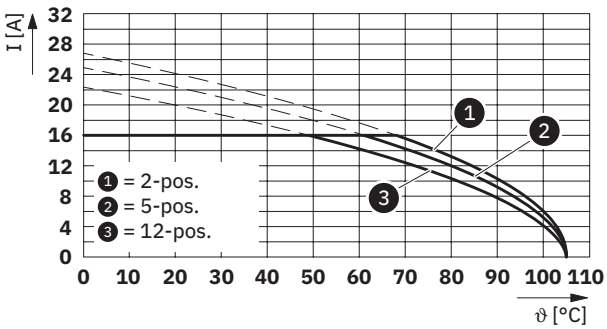
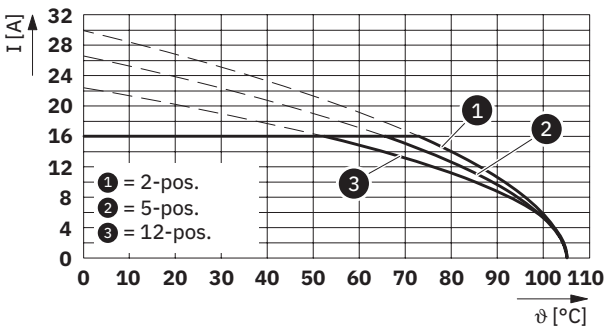
19 Current carrying capacity/derating curves

Specification	IEC 61984:2008-10
Note	Representation based on IEC 60512-5-2:2002-02
Note	For number of positions, see diagram
Reduction factor	0.8
Conductor cross section	2.5 mm²

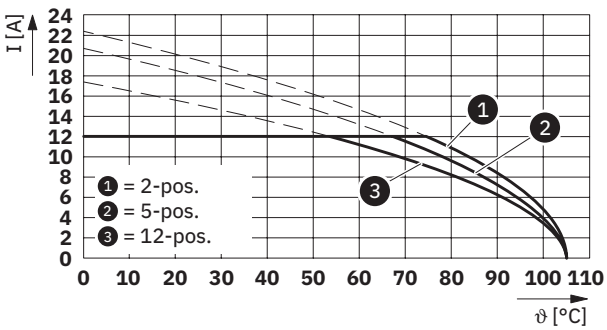
Type: XPC 2,5/...-ST-5,08 with CCA 2,5/...-G-5,08 P...THR



Type: XPC 2,5/...-ST-5,08 with CCVA 2,5/...-G-5,08 P...THR



Type: XPC 2,5/...-ST-5,08 with MSTBVA 2,5/...-G-5,08



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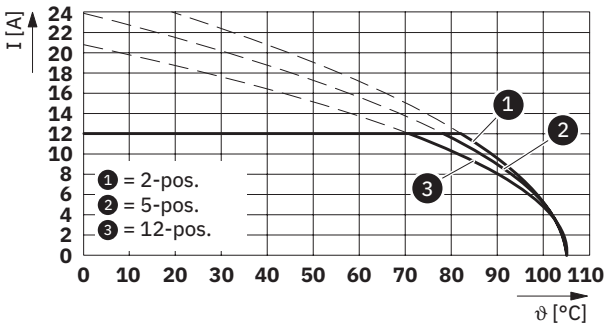
1491342 XPC 2,5/ 3-ST-5,08

Type: XPC 2,5/...-ST-5,08 with FKIC 2,5/...-ST-5,08



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20 Environmental and durability tests

20.1 Vibration test

Specification	IEC 60068-2-6:2007-12
Result	Test passed
Frequency	10 - 150 - 10 Hz
Sweep speed	1 octave/min
Amplitude	0.35 mm (10 Hz ... 60.1 Hz)
Acceleration	50 m/s^2 (60.1 Hz ... 150 Hz)
Test duration per axis	2.5 h
Test directions	X-, Y- and Z-axis
Note	

20.2 Insulation resistance

Specification	IEC 60512-3-1:2002-02
Result	Test passed
Insulation resistance, neighboring positions	> 5 MΩ

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21 Type approval and special tests

22 Classification for connectors

Specification	IEC 61984:2008-10
Main features	Connectors without switching capacity (COC)
	Fixed connectors
Strain relief elements	without strain relief
Connection method	Can be reconnected
Protection against electric shock	Not encapsulated - touch-proof when inserted
Protective conductor	without PE
Locking	no
Connection method	Screwless terminal points

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23 Approvals / Certificates

cULus Recognized	Voltage [V]	Current [A]	Cross section [AWG]	Cross section [mm²]
Usegroup B				
Standard	300 V	16 A	20 - 12	-
Usegroup F				
USR application only	250 V	16 A	20 - 12	-
Usegroup D				
Alternative 1	300 V	10 A	20 - 12	-

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24 Commercial Data

Item no.	1491342
Type	XPC 2,5/ 3-ST-5,08
Packing unit	100
Net weight	2.22 g
GTIN	4063151937997
	Information that applies locally, see link on page 1

25 corresponding headers

Item no.	Type
1954391	CC 2,5/ 3-G-5,08 P26THR
1954922	CCA 2,5/ 3-G-5,08 P26THR
1955390	CCV 2,5/ 3-G-5,08 P26THR
1955866	CCVA 2,5/ 3-G-5,08 P26THR
1762075	MDSTB 2,5/ 3-G-5,08
1762376	MDSTB 2,5/ 3-G1-5,08
1842076	MDSTBA 2,5/ 3-G-5,08
1763087	MDSTBV 2,5/ 3-G-5,08
1736742	MDSTBV 2,5/ 3-G1-5,08
1845345	MDSTBVA 2,5/ 3-G-5,08
1802414	MDSTBW 2,5/ 3-G-5,08
1759020	MSTB 2,5/ 3-G-5,08
1757255	MSTBA 2,5/ 3-G-5,08
1758021	MSTBV 2,5/ 3-G-5,08
1755749	MSTBVA 2,5/ 3-G-5,08
1735879	MSTBW 2,5/ 3-G-5,08
1767384	SMSTBA 2,5/ 3-G-5,08

26 Accessories

Description	Item No.	**Type**
Coding profile, is inserted into the slot on the plug or inverted header, red insulating material	1734634	CP-MSTB
Actuation tool, for ST terminal blocks, insulated, also suitable for use as a bladed screwdriver, size: 0.6 x 3.5 x 100 mm, 2-component grip, with non-slip grip	1205053	SZS 0,6X3,5
	1020292	MPS-MT 1-S4-B RD VPE1
Crimping pliers, for ferrules without insulating collar according to DIN 46228 Part 1 and ferrules with insulating collar according to DIN 46228 Part 4, 0.25 mm^2 ... 6.0 mm^2, lateral entry, trapezoidal crimp	1212034	CRIMPFOX 6

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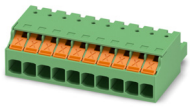
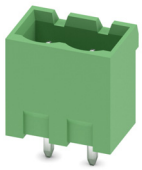
27 Combination tests

				
XPC 2,5/..-ST	**CCA 2,5/..-G**	**CCVA 2,5/..-G**	**MSTBVA 2,5/..-G**	**FKIC 2,5/..-ST**
IEC 61984	IEC 61984	IEC 61984	IEC 61984	IEC 61984
Mechanical tests (A)				
Insertion/withdrawal force per position	approx. 9 N / 8 N	approx. 9 N / 8 N	approx. 9 N / 7 N	approx. 9 N / 6 N
Polarization when inserted Requirement >20 N	Test passed	Test passed	Test passed	Test passed
Contact holder in insert Requirements >20 N	Test passed	Test passed	Test passed	Test passed
Durability tests (B)				
Contact resistance R_1 1st level	1.2 mΩ	1.2 mΩ	2.2 mΩ	1.6 mΩ
Contact resistance R_1 2nd level				
Insertion/withdrawal cycles	25	25	25	25
Contact resistance R_2	1.2 mΩ	1.2 mΩ	2.3 mΩ	1.5 mΩ
Rated impulse voltage at sea level Voltage waveform ≥ (1.2/50 µs)	4.8 kV	4.8 kV	4.8 kV	4.8 kV
Power-frequency withstand voltage Voltage waveform ≥ (50/60 Hz)	2.21 kV	2.21 kV	2.21 kV	2.21 kV
Insulation resistance Requirements > 5 MΩ	> 5 MΩ	> 5 MΩ	> 5 MΩ	> 5 MΩ
Thermal tests (C)				
Tested number of positions	12	12	12	12
Tested conductor cross section	2.5 mm^2	2.5 mm^2	2.5 mm^2	2.5 mm^2
Test current	16 A	16 A	12 A	12 A
Upper limiting temperature Requirements < 100°C	Test passed	Test passed	Test passed	Test passed
Climatic tests (D)				
Test sequence 1: low temperature storage	-40 °C/2 h	-40 °C/2 h	-40 °C/2 h	-40 °C/2 h
Test sequence 2: heat storage	105 °C/168 h	105 °C/168 h	105 °C/168 h	105 °C/168 h
Test sequence 3: noxious gas storage	0.2 dm^3 SO_2 on 300 dm^3/ 40 °C/1 cycle	0.2 dm^3 SO_2 on 300 dm^3/ 40 °C/1 cycle	0.2 dm^3 SO_2 on 300 dm^3/ 40 °C/1 cycle	0.2 dm^3 SO_2 on 300 dm^3/ 40 °C/1 cycle
Rated impulse voltage at sea level Voltage waveform ≥ (1.2/50 µs)	4.8 kV	4.8 kV	4.8 kV	4.8 kV
Power-frequency withstand voltage Voltage waveform ≥ (50/60 Hz)	2.21 kV	2.21 kV	2.21 kV	2.21 kV
Environmental and endurance tests (E)				
Specification	IEC 61984:2008-10	IEC 61984:2008-10	IEC 61984:2008-10	IEC 61984:2008-10
Degree of protection	Finger safety with IP20 test finger	Finger safety with IP20 test finger	Finger safety with IP20 test finger	Finger safety with IP20 test finger

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